Robinson Vidva, MS, MBA

Biomedical Data Scientist

Washington, DC area · hello@robinsonvidva.com

6 patents (3 granted: 2 US, 1 European) | 7 peer-reviewed publications, 2 preprints, 14 abstracts | 150+ citations

SUMMARY

Biomedical data scientist with 18 years applying multi-omics, machine learning, and NGS analysis across computational oncology, immuno-oncology, drug discovery, and pediatric neurodevelopment, with a focus on integrating mutational and gene-expression data to characterize biomarkers of therapeutic response and on linking molecular pathways to organism-level outcomes.

Career spans industry computational drug discovery (Cellworks Life), independent consulting (Digirobi Solutions), and current translational pediatric research (Children's National Hospital), including leadership of 10+ scientist teams and institutional standards for model validation.

RESEARCH FOCUS

Computational oncology, immuno-oncology, and immunology; cancer precision medicine; cancer genomics and NGS data analysis; digital drug-response modeling; cancer therapeutics and drug repurposing; hematologic malignancies (AML); biomarkers of therapeutic response; systems-biology pathway modeling; pediatric and neonatal translational research; preterm neurodevelopmental injury (NEC, SIDS); behavioral phenotyping with physical therapy intervention; multi-domain clinical data analysis (MRI, ultrasound, neurodevelopmental assessments); open-source biomedical infrastructure (MyVivarium); autoimmune and inflammatory disease (rheumatoid arthritis, Crohn's); emerging pediatric neuro-oncology (neuroblastoma, astrocytoma); immune-modulated therapeutics.

PROFESSIONAL EXPERIENCE

Research Technician — March 2025 – Present

Children's National Hospital — Kratimenos Lab, Sheikh Zayed Institute for Pediatric Surgical Innovation, Washington, DC

- Build and maintain reproducible RNA-seq / NGS analysis pipelines for multi-omics analyses (RNA-seq, metabolomics, genomics) of mouse and human datasets in Python, R, and SQL to identify differentially expressed genes, pathways, and GO terms underlying preterm cerebellar deficits in necrotizing enterocolitis (NEC), sudden infant death syndrome (SIDS), and related neurodevelopmental disorders.
- Quantify behavioral data from 250+ mice using DeepLabCut and B-SOiD to characterize the effects of physical therapy and environmental enrichment on preterm neurodevelopmental outcomes, including three-chamber sociability assays.
- Develop statistical and machine-learning predictive models on Erasmus Ladder motor-coordination data from 160 preterm mice to identify gait and cerebellar deficits.
- Conduct longitudinal statistical analysis of multi-domain clinical/medical-record data from 200+ preterm infants (MRI, cranial ultrasound, Fenton growth charts, and neurodevelopmental scores) to characterize NEC developmental trajectories.

Founder & Lead Computational Biologist September 2019 – June 2022

Digirobi Solutions — Bengaluru, Karnataka, India

- Founded an independent computational biology consultancy supporting 5+ engagements (3 commercial, 2 academic) in cancer therapeutics and open-source scientific tooling.
- Developed cancer genomics, drug, and nutrition databases through targeted data mining of open and proprietary sources.
- Modeled the SRC kinase signaling pathway in MATLAB; contributed to a peer-reviewed publication on molecular signaling.
- Analyzed cancer nutrigenomics data; co-invented a nutraceutical combination therapy for cancer, granted in the US (11,478,444 B2) and Europe (EP 3,861,992 B1).
- First-authored MyVivarium, an open-source animal colony management platform deployed at 3 partner laboratories (200+ animals), published in Computational and Structural Biotechnology Journal (2025).

Senior Lead Scientist November 2017 – September 2019

Cellworks Life — Bengaluru, Karnataka, India

- Led a multidisciplinary research team of 10+ scientists developing computational precision-medicine platforms for oncology; defined institutional standards for modeling reproducibility and validation.
- Designed and validated digital drug-response models achieving 80%+ correlation with experimental data across multiple solid and hematologic cancer types, integrating TCGA, MSK-IMPACT, and 10+ proprietary genomic, transcriptomic, and pharmacological datasets to characterize biomarkers of response to oncology treatments.
- Constructed a 500+ cancer cell-line database with associated IC50 drug-sensitivity profiles, supporting systematic validation of model predictions and contributing to peer-reviewed publications.
- Curated 15+ external clinical datasets used for training, validation, and benchmarking of predictive models, supporting multiple publications and conference abstracts.

Lead Scientist January 2014 – October 2017

Cellworks Life — Bengaluru, Karnataka, India

- Developed an integrated drug knowledge repository of ~200 compounds, cataloging target profiles, mechanisms of action, pharmacokinetic parameters, and resistance/activity biology to support therapeutic discovery.
- Translated computational findings into peer-reviewed publications and intellectual property; contributing inventor on US 10,098,880 B2 (PTEN/TP53-targeted cancer combination therapy), with responsibility for claim drafting and patent prosecution.
- Evaluated computational therapeutic candidates and contributed to collaborative research with pharmaceutical partners.
- Mentored junior scientists and led 25+ training sessions on computational biology and data analytics.

Associate Lead Scientist June 2008 – December 2013

Cellworks Life — Bengaluru, Karnataka, India

- Developed a systems-biology model of rheumatoid arthritis incorporating 8 distinct cell types with comprehensive in vivo validation.
- Identified 2 synergistic drug combinations through systematic computational screening; advanced candidates for clinical collaboration.
- Contributions resulted in 2 patent applications and presentations at the American College of Rheumatology; collaborative work on inflammatory skin disorders yielded peer-reviewed publications.
- Established institutional quality standards for computational model development and validation.

Associate Biomodeling Scientist July 2007 – May 2008

Cellworks Life — Bengaluru, Karnataka, India

- Constructed mechanistic models of immunological signaling in macrophages, incorporating 20+ interconnected pathways and gene-regulatory networks.
- Developed systematic validation frameworks through perturbation analyses, establishing foundational methodology for the organization's computational biology platform.

TECHNICAL SKILLS

Programming languages — Python, R, SQL, MATLAB, C/C++, Unix shell, HTML/CSS/JavaScript, PHP.

Software & web application development — Full-stack web application development; software engineering best practices including version control (Git), issue tracking (GitHub, Jira), and CI/CD pipelines; workflow management (Snakemake), containerization (Docker, Singularity), and reproducible analysis pipelines; database design; server and Linux/Unix administration; domain and DNS management; high-performance and cloud computing (LSF job scheduling, AWS); IoT integration for research environments.

Bioinformatics & multi-omics data analysis — Analysis of bulk and single-cell RNA-seq, metabolomics, genomics, and spatial transcriptomics data; NGS DNA and RNA sequencing analysis; variant / mutational analysis (SNV, CNV); pathway and cell–cell communication analysis; multi-domain clinical dataset integration (MRI, ultrasound, neurodevelopmental scoring).

Clinical data & compliance — Analysis of de-identified clinical and medical-record data under IRB-approved protocols; HIPAA/PHI-compliant handling and familiarity with data-use agreements.

Computational & mathematical modeling — Mechanistic network modeling; ODE-based and stochastic pathway modeling; systems biology; predictive modeling and machine learning; generalized linear models, longitudinal/repeated-measures (mixed-effects) models, and survival analysis.

Quantitative behavioral phenotyping — Behavioral data analysis using DeepLabCut, B-SOiD, and BONSAI; quantitative analysis of three-chamber sociability and Erasmus Ladder readouts; motor and gait phenotyping.

Visualization & analytics — Tableau, matplotlib, seaborn, ggplot2; scientific reporting and dashboard development.

Machine learning & deep-learning frameworks — scikit-learn, TensorFlow, Keras, PyTorch; applied to predictive modeling and pose-estimation/behavioral classification (DeepLabCut, B-SOiD).

EDUCATION

Master of Science (M.S.), Data Analytics July 2024 – January 2025
Western Governors University, Salt Lake City, UT, USA

Master of Business Administration (M.B.A.), Project Management November 2012 – February 2020
Sikkim Manipal University (Distance Education), Gangtok, Sikkim, India

Bachelor of Technology (B.Tech.), Biotechnology July 2003 – April 2007
Bharathidasan University, Tiruchirappalli, Tamil Nadu, India

PEER-REVIEWED PUBLICATIONS

1. Giannopoulos K, Karikis I, Byrd C, Sanidas G, Wolff N, Triantafyllou M, Simonti G, **Vidva R**, Koutroulis I, Theocharis S, Kratimenos P. Eph/Ephrin-Mediated Immune Modulation: A Potential Therapeutic Target for Skin Cancer. Frontiers in Immunology. 2025;16:1539567. doi:10.3389/fimmu.2025.1539567

2. **Vidva R**, Raza MA, Prabhakaran J, Sheikh A, Sharp A, Ott H, Moore A, Fleisher C, Netherton H, Goldstein E, Pitychoutis PM, Nguyen TV, Sathyanesan A. MyVivarium: A cloud-based lab animal colony management application with real-time ambient sensing. Computational and Structural Biotechnology Journal. 2025;27:612–623. doi:10.1016/j.csbj.2025.01.025

3. Kratimenos P, Vij A, **Vidva R**, Koutroulis I, Delivoria-Papadopoulos M, Gallo V, Sathyanesan A. Computational analysis of cortical neuronal excitotoxicity in a large animal model of neonatal brain injury. Journal of Neurodevelopmental Disorders. 2022;14(1):26. doi:10.1186/s11689-022-09431-3

4. Fischer CL, Bates AM, Lanzel EA, Guthmiller JM, Johnson GK, Singh NK, Kumar A, **Vidva R**, Abbasi T, Vali S, Xie XJ, Zeng E, Brogden KA. Computational Models Accurately Predict Multi-Cell Biomarker Profiles in Inflammation and Cancer. Scientific Reports. 2019;9(1):10877. doi:10.1038/s41598-019-47381-4

5. Drusbosky LM, **Vidva R**, Gera S, Lakshminarayana AV, Shyamasundar VP, Agrawal AK, Talawdekar A, Abbasi T, Vali S, Tognon CE, Kurtz SE, Tyner JW, McWeeney SK, Druker BJ, Cogle CR. Predicting response to BET inhibitors using computational modeling: A BEAT AML project study. Leukemia Research. 2019;77:42–50. doi:10.1016/j.leukres.2018.11.010

6. Borgwardt DS, Martin AD, Van Hemert JR, Yang J, Fischer CL, Recker EN, Nair PR, **Vidva R**, Chandrashekaraiah S, Progulske-Fox A, Drake D, Cavanaugh JE, Vali S, Zhang Y, Brogden KA. Histatin 5 binds to Porphyromonas gingivalis hemagglutinin B (HagB) and alters HagB-induced chemokine responses. Scientific Reports. 2014;4(1):3904. doi:10.1038/srep03904

7. Harvey LE, Kohlgraf KG, Mehalick LA, Raina M, Recker EN, Radhakrishnan S, Prasad SA, **Vidva R**, Progulske-Fox A, Cavanaugh JE, Vali S, Brogden KA. Defensin DEFB103 bidirectionally regulates chemokine and cytokine responses to a pro-inflammatory stimulus. Scientific Reports. 2013;3(1):1232. doi:10.1038/srep01232

PREPRINTS

1. Sanidas G, Simonti G, Ghaemmaghami J, Woyshner K, Wolff N, Byrd C, Triantafyllou M, Lowe C, Salisbury H, Goldstein E, Sathyanesan A, **Vidva R**, Koutroulis I, Stein-O'Brien G, Sidiropoulos DN, Gallo V, Kratimenos P. Prematurity Insults Remodel Cerebellar Development and Behavior. bioRxiv. 2025. doi:10.1101/2025.07.22.664624
2. Sanidas G, Simonti G, Ghaemmaghami J, Woyshner K, **Vidva R**, Wolff N, Triantafyllou M, Lowe C, Vij A, Pettersen HS, Chandereng T, Koutroulis I, Stein-O'Brien G, Sidiropoulos DN, Gallo V, Kratimenos P. Intrinsic gestational timing governs human cerebellar development after preterm birth. bioRxiv. 2025. doi:10.1101/2025.09.01.673244

PATENTS

Granted

1. Vali S, Abbasi T, Stopka T, Minarik L, Singh NK, Usmani S, Radhakrishnan S, Sikora H, **Vidva R**, Pimkova K. Use of scientifically matched plant supplements combined with antineoplastic compounds for the treatment of hematological malignancies. US Patent No. 11,478,444 B2 (issued October 25, 2022). Assignee: Brio Ventures LLC.
2. Vali S, Abbasi T, Stopka T, Minarik L, Singh NK, Usmani S, Radhakrishnan S, Sikora H, **Vidva R**, Pimkova K, Padouk JM, Yilma P. Combination therapy comprising plant derived compounds and antineoplastic compounds for treatment of hematological malignancies. European Patent No. EP 3,861,992 B1 (granted September 6, 2023). Assignee: Cannafamily AS. European counterpart of US 11,478,444 B2.
3. Vali S, Usmani S, Sultana Z, Kumar A, Abbasi T, **Vidva R**. Combination of nelfinavir, metformin and rosuvastatin for treating cancer caused by aberrations in PTEN/TP53. US Patent No. 10,098,880 B2 (issued October 16, 2018). Assignee: Cell Works Group Inc.

Published Applications

1. Vali S, Abbasi T, Fernandes P, Rajagopalan S, Alam A, **Vidva R**, Nair PR, Kapoor S, Radhakrishnan S, Sultana Z, Ramanujan KS, Tiwari KK, Kumar A, Singh NK, Agrawal AK, Talawdekar AA, Usmani S, Singh R. System and method for development of therapeutic solutions. US Patent Application Publication No. 2015/0302167 A1 (published October 22, 2015). Assignee: Cell Works Group Inc.
2. Vali S, **Vidva R**, Nair PR, Fernandes P, Abbasi T, Radhakrishnan S. Compositions, process of preparation of said compositions and method of treating inflammatory diseases. US Patent Application Publication No. 2015/0098993 A1 (published April 9, 2015). Assignee: Cell Works Group Inc.
3. Vali S, **Vidva R**, Nair PR, Fernandes P, Abbasi T, Radhakrishnan S. Compositions, process of preparation of said compositions and method of treating inflammatory diseases. US Patent Application Publication No. 2014/0127295 A1 (published May 8, 2014). Assignees: Cellworks Research India Pvt Ltd; Cell Works Group Inc.

CONFERENCE PRESENTATIONS

1. Sanidas G, Lowe C, Byrd C, **Vidva R**, Simonti G, Chandereng T, Gallo V, Kratimenos P. Necrotizing Enterocolitis Selectively Disrupts Pontine Development in the Preterm Brain. Society for Neuroscience (SfN) 2025. Poster #PSTR103.21.

2. Simonti G, **Vidva R**, Sanidas G, Byrd C, Lowe C, Triantafyllou M, Wolff N, Chandereng T, Koutroulis I, Gallo V, Kratimenos P. Early postnatal rehabilitation mitigates motor, growth, and social deficits in perinatal brain injury. Society for Neuroscience (SfN) 2025. Poster #PSTR019.10.
3. Cogle CR, Abbasi T, Singh NK, Sauban M, Raman RK, **Vidva R**, et al. AraC-Daunorubicin-Etoposide (ADE) Response Prediction in Pediatric AML Patients Using a Computational Biology Modeling (CBM) Based Precision Medicine Workflow. Blood. 2018;132:4034. doi:10.1182/blood-2018-99-115775
4. Tyner JW, Druker BJ, Tognon CE, Kurtz SE, Drusbosky LM, Sahu D, **Vidva R**, et al. Predicting Response to Dasatinib Using a Computational Model and Its Validation: A Beat AML Project Study. Blood. 2018;132:1541. doi:10.1182/blood-2018-99-115678
5. Tyner JW, Druker BJ, Kurtz SE, Tognon CE, Drusbosky LM, **Vidva R**, et al. Predicting Response to BET Inhibitor in Combination with Palbociclib/Sorafenib Using a Computational Model: A Beat AML Project Study. Blood. 2018;132:1540. doi:10.1182/blood-2018-99-115284
6. Drusbosky LM, Turcotte M, Castillo P, **Vidva R**, et al. Predicting Response to CDK4/6 Inhibitors and Combinations Using a Computational Biology Model: A Beat AML Project Study. Blood. 2017;130:3909.
7. Drusbosky LM, **Vidva R**, et al. Predicting Response to BET Inhibitors Using a Computational Model and Its Validation: A Beat AML Project Study. Blood. 2017;130:3908.
8. Drusbosky LM, **Vidva R**, et al. Predicting Response to IDH1/IDH2 Inhibitors Beyond IDH Mutations Using a Computational Model: A Beat AML Project Study. Blood. 2017;130:3907.
9. Drusbosky L, Abbasi T, Vali S, Radhakrishnan S, Singh NK, Usmani S, Parashar D, **Vidva R**, et al. A Genomic Signature Predicting Venetoclax Treatment Response in AML. Blood. 2016;128(22):1713. doi:10.1182/blood.V128.22.1713.1713
10. Fischer C, Abbasi T, Vali S, Nair P, **Vidva R**, Tiwari K, Brogden KA. A Predictive Model of an Oral Inflammatory Response. Journal of Dental Research. 2014;93(Spec Iss A):1369.
11. Harvey L, Recker E, Raina M, Radhakrishnan S, Prasad S, **Vidva R**, Cavanaugh J, Vali S, Brogden KA. HBD3 Both Enhances, Attenuates Dendritic Cell MAPK Responses to HagB. Journal of Dental Research. 2013;92(Spec Iss A):2820.
12. Singh G, **Vidva R**, Nair P, Radhakrishnan S, Fernandes P, Abbasi T, Refino C, Cruz JD, Vali S. Predictive Software-Based Mathematical Modeling: A Novel Approach to Development of Oral Therapies for Rheumatoid Arthritis – Validation in a Murine Collagen Induced Arthritis Model. Arthritis and Rheumatism. 2012;64(10 Suppl):S149–S150. doi:10.1002/art.37735
13. Vali S, Refino C, Cruz JD, **Vidva R**, Nair P, Radhakrishnan S, Fernandes P, Abbasi T, Singh G. Novel Combination Therapy of Existing Repurposed Therapies Designed by Predictive Software Modeling Shows Profound Impact on Disease Progression in a Murine Collagen Induced Arthritis Model. Arthritis and Rheumatism. 2012;64(10 Suppl):S148. doi:10.1002/art.37735
14. Vali S, Hegedus C, Bhattacharjee C, **Vidva R**, Schmelzer K, Inceoglu B, Hammock BD. Insights into the Anti-inflammatory Action of the Soluble Epoxide Hydrolase Inhibitor through Systems Biology based in silico Modeling Approach. FASEB Journal. 2008;22(1_supplement):479.13. doi:10.1096/fasebj.22.1_supplement.479.13

SOFTWARE & OPEN-SOURCE CONTRIBUTIONS

MyVivarium — Open-source cloud-based lab animal colony management platform with real-time ambient sensing. First-author publication (Computational and Structural Biotechnology Journal, 2025); deployed at 3 partner laboratories supporting 200+ animals. Ongoing development to support diverse experimental, husbandry, and operational laboratory needs. github.com/myvivarium/MyVivarium

MENTORING

Mentored 10+ junior scientists and engineers across Cellworks Life and Digirobi Solutions; designed and delivered 25+ training sessions in computational biology, data analytics, and modeling methodology.

CERTIFICATIONS

Cancer Biology Specialization — Johns Hopkins University (Coursera)

Drug Development Product Management Specialization — University of California San Diego (Coursera)

DL-101 General Course on Intellectual Property — World Intellectual Property Organization (WIPO)

DL-204 Biotechnology and Intellectual Property — World Intellectual Property Organization (WIPO)

Advanced Data Modeling, Data Analytics Fundamentals, Data Preparation — Western Governors University

Demystifying Biomedical Big Data: A User's Guide — Georgetown University (edX)

Genomic and Precision Medicine — University of California San Francisco (Coursera)

Design and Interpretation of Clinical Trials — Johns Hopkins University (Coursera)

PROFESSIONAL MEMBERSHIPS

International Society for Computational Biology (ISCB), 2024 – Present.

LANGUAGES

English (Fluent), Tamil (Native), Kannada (Conversational).

ONLINE IDENTIFIERS

Website — robinsonvidva.com · ORCID — orcid.org/0000-0002-2604-5448 · LinkedIn — linkedin.com/in/robinson-vidva · GitHub — github.com/robinson-vidva

REFERENCES

Professional references are available on request.